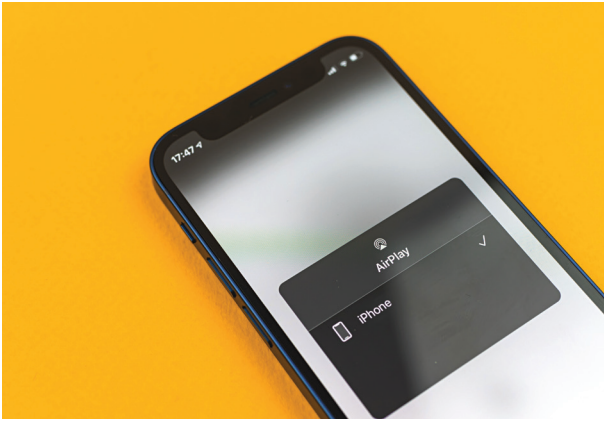




- **Chromecast built-in / Google Home groups** – create persistent speaker groups in the Google Home app and cast to them from Android apps or Chrome; Google’s help pages detail creating and editing groups.
- **Alexa Multi-Room Music** – the Alexa app lets you create named groups (e.g., “Upstairs”); Amazon also auto-creates an Everywhere group when you have more than one Echo.
- **Spotify Connect** – not a sync protocol by itself, but lets the Spotify app control playback directly on devices (speakers/AVRs/TVs) over your network. It’s wonderfully robust, but one account only plays on one device at a time.
- **Sonos** – a whole ecosystem; grouping rooms is built into the app and even the hardware (press-and-hold Play/Pause to group with whatever’s playing).
- **HEOS (Denon/Marantz)** – HEOS Built-in speakers, AVRs and soundbars group easily; TV Sound Grouping can distribute TV audio to other HEOS rooms (with adjustable delay if you hear echo).
- **Yamaha MusicCast** – add up to 32 devices in the app, link up to 10 at once for simultaneous playback.
- **DTS Play-Fi** – brand-agnostic high-res, sub-millisecond sync ecosystem across multiple manufacturers.
- **Roon (RAAT)** – for enthusiasts with a library (local or streaming). Roon’s RAAT is built for bit-perfect, synchronized, multi-room playback with easy grouping.

Rule of thumb: If your home is Apple-heavy, AirPlay 2 + HomePods/compatible speakers is simplest. Mixed Android/TV ecosystem? Chromecast groups or Alexa MRM. Audiophile/AVR-centric? HEOS/MusicCast/Play-Fi/Roon.

2) Design your zones (rooms) in 15 minutes

1. **List the spaces** where you want audio now (Living, Kitchen, Balcony) and later (Bedrooms).
2. **Pick one “primary” protocol** your family will actually use daily (AirPlay 2, Chromecast, Alexa MRM, Sonos, etc.).
3. **Map devices:**
 - TV rooms → soundbar/AVR with your protocol (e.g., Denon

AVR with HEOS; Sonos Arc; Yamaha AVR with MusicCast). HEOS and Sonos both support grouping TV sound to other rooms – HEOS exposes a specific “TV Sound Grouping” toggle and synchronization control.

- Utility rooms/balconies → small smart speakers.

4. **Account hygiene:** If two different rooms must play different Spotify streams at once, you’ll want separate Spotify accounts (e.g., Family plan) because one account plays on one device at a time.

3) Solid networking = stable sync

Multi-room lives and dies on Wi-Fi. Get these right before you blame your speakers.

- **Router & mesh:** If your flat/home is large or has brick walls, use a mesh system; for best reliability, wire the nodes using Ethernet backhaul (mesh vendors and networking guides recommend it to improve stability/throughput).
- **Wi-Fi 6/6E/7:** Newer routers cope better with many devices. Wi-Fi 7 adds Multi-Link Operation (MLO) so clients can use two bands at once for lower, more predictable latency – useful when several rooms are syncing.
- **WMM (Wi-Fi Multimedia):** Leave it enabled on the router – Apple’s guidance and enterprise docs note WMM prioritizes voice/video streams so music doesn’t stutter when someone starts a large download.
- **Sonos specifics:** Sonos can run on your Wi-Fi or its own



SonosNet when one unit (or a Boost) is wired – useful in tricky RF environments; Sonos documents how to check or change modes and tweak channels.

- **Roon note:** Roon’s own guide calls mesh “a middle-ground that covers most homes” and praises products that expose Ethernet at satellites – great for wiring an AVR/streamer where there’s no LAN port nearby.
- **Quick checks:** Place nodes sensibly (not in cabinets), keep 2.4 GHz for range and 5/6 GHz for throughput, and wire what you can (router → TV/AVR) to reduce Wi-Fi chatter for everything else.